

RESILIENT & TRUSTWORTHY ONLINE ASSESSMENT ECOSYSTEM

ASTRA

Assessment Security, Telemetry & Resilience Architecture

Failures pause. Exams don't.

Power cuts and server crashes no longer cancel computer-based exams.
No answer lost. No minute lost. A tamper-proof record of everything.

THE PROBLEM

Computer-based exams **collapse** over ordinary faults

489

centres lost a CUET-UG shift in one day: the paper arrived late (2022)

2 days

in a row, the same Jaipur centre failed NEET-PG, then an AIIMS exam (2026)

~3 hrs

MPESB candidates waited before a shift was cancelled for a server failure (2026)

28 lakh

SSC CGL 2025 candidates hit by server crashes and cancelled centres

MADHYA PRADESH

MPESB has faced every kind of digital failure

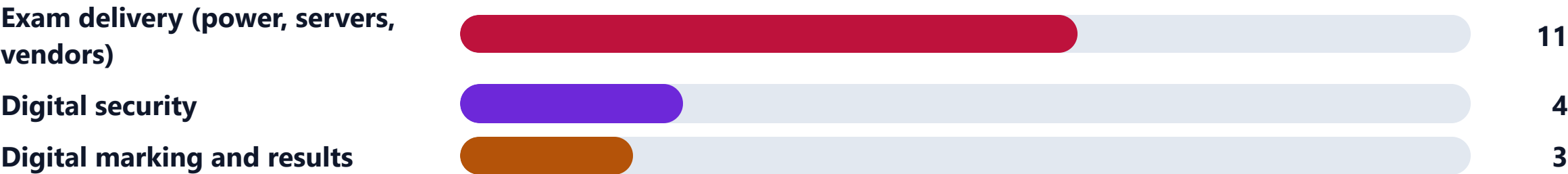
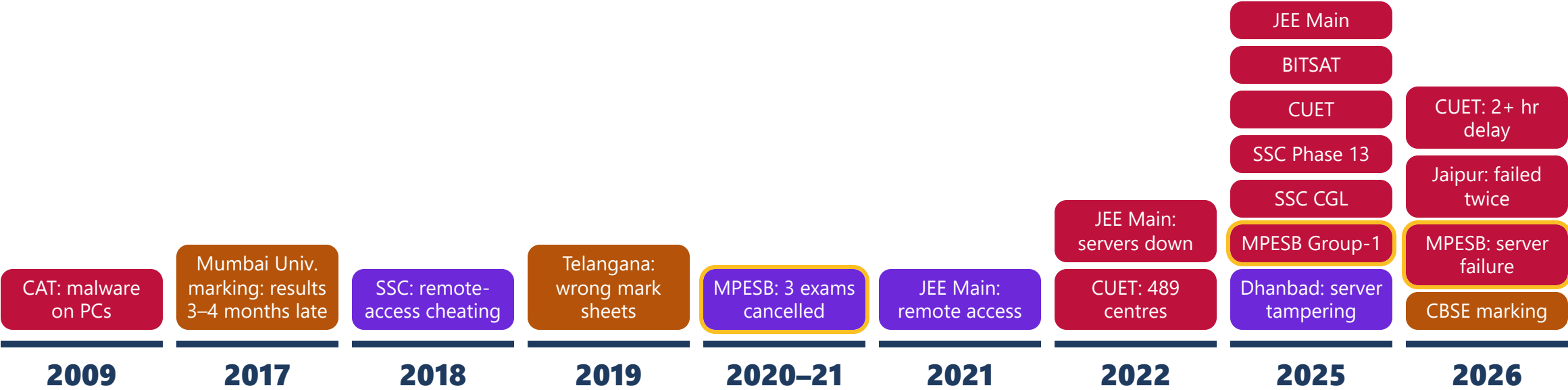
2020–21 · Security: 3 exams cancelled after papers were downloaded from unauthorised computers

May 2025 · Technical: a shift postponed with 4,818 candidates already at 42 centres

June 2026 · Server: the vendor's server failed; candidates waited ~3 hours; shift cancelled

Today there is only one remedy: **cancel and re-test**. Candidates lose money, time and trust.

18 digital failures, 2009 – 2026: it keeps happening



Yellow outline = Madhya Pradesh. Only digital and technical failures are counted (no paper leaks).

Four challenges ASTRA solves

1. Exams get cancelled

One power cut, server crash or late paper stops entire shifts. Candidates lose answers and time.

2. Problems are spotted too late

No early warning, no live view of centres, and candidates wait hours for an official update.

3. Decisions are ad hoc and unfair

Whether to continue or re-test, and how much time to give back, is decided case by case.

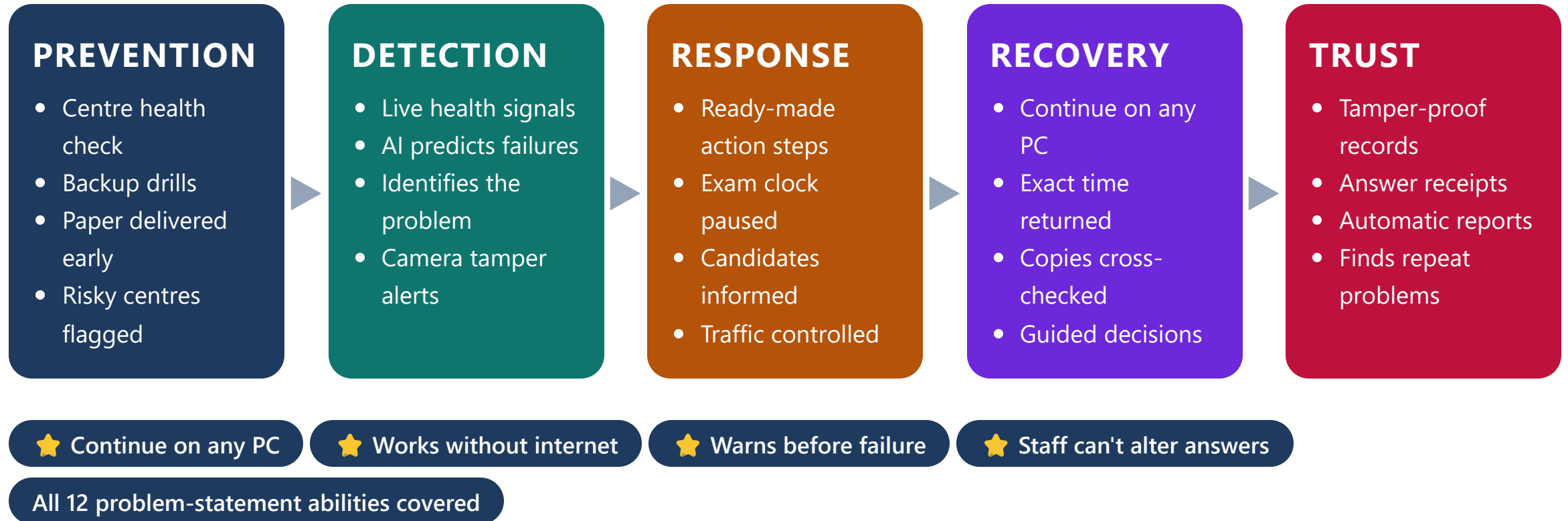
4. Records can't be trusted

Insiders can tamper with data, evidence comes late, and the same failures keep repeating.

The root cause: exams depend on everything working at once, nobody sees problems early, and no system can restore a candidate's exam exactly as it was.

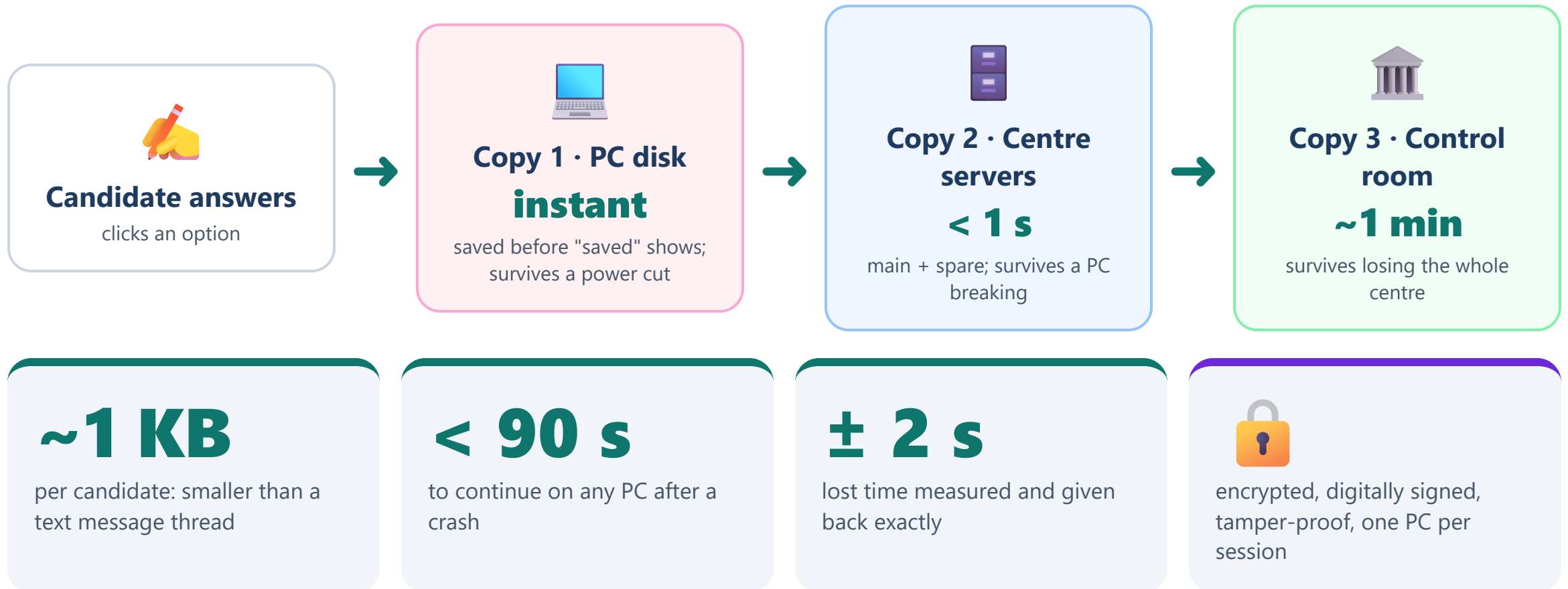
ASTRA protects the **whole exam**, not just the computer

A resilience and trust layer for computer-based exams. It works across the entire exam lifecycle:



The answer backup ("Capsule"): every answer, in 3 places

A tiny, locked backup of each candidate's exam (answers, question order, position and time left), saved on every click.



Three layers. Each survives the failure of the next.

1

Exam PCs

at every seat

- Locked-down Linux
- Offline exam app
- Answers saved to disk
- No internet access

↕ private, encrypted exam network

2

Centre server

main + spare, auto takeover

- Official exam clock
- Answer backups
- Action steps + alerts
- Fast AI (offline)
- Centre dashboards
- CCTV recorder
- Own UPS, separate power

↕ secure one-way link · calls out only · ~1/min · can be switched off

3

Control room

main + backup site

- Exam-authority dashboard
- Decision help
- Messages
- AI assistant (local)
- Risk reports
- Evidence store



Answer backup (Capsule)

Copy 1 · PC disk
instant

Copy 2 · Centre servers
< 1 second

Copy 3 · Control room
~1 minute



Encrypted



Digitally signed



Tamper-proof chain



One PC per session

 **Resilience rule: PCs keep working without the centre server; centres keep working without the internet or control room.**

WHEN SOMETHING FAILS

From failure to recovery in under 2 minutes



Problem	What ASTRA does	Exam stops?
⚡ Room power cut	Pause clocks, inform everyone, continue when power returns	Paused, not cancelled
💻 One PC fails	Candidate moves to a spare PC with the same answers and time	No
🖨 Main centre server fails	Spare server takes over automatically	No
🌐 Internet or vendor server down	Centre runs on its own; data syncs later	No
🔑 Paper key late	Backup key holders unlock it; start shifts equally for all	No
🏠 Centre unusable	Candidates move to another centre the same day and continue	Moved, not cancelled

SEE IT WORKING

Design prototype: a power cut, from five points of view

Candidate: "answers saved, clock paused"

Centre admin: incident steps done automatically

Exam authority: decision with 2 approvals

Continue: 0 minutes lost

AI: risk 8% → 71% before the failure

Proof: tampering detected live



Try it: yash-kasera.github.io/ASTRA/prototype
Clickable, simulated data · fingerprints calculated live in the browser

Two layers of AI. The exam never waits for AI.

Fast AI: inside the centre

Works offline · answers in milliseconds · typed decisions only

 Failure predictor	Breakdown chance in next 5–15 min, with reasons
 Problem sorter	Type and seriousness of each incident
 Report reader	Hindi / Hinglish invigilator notes → alerts
 Unusual activity	e.g. answers with no keyboard or mouse use
 Camera checks	Covered, moved or looping cameras (computer vision)

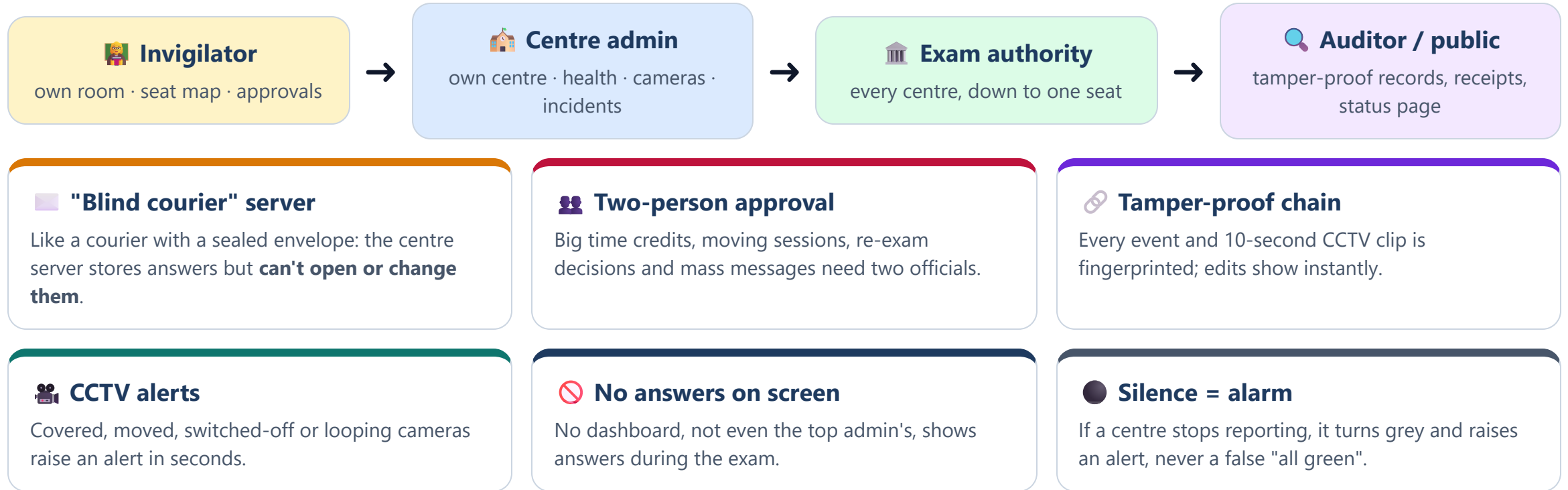
Assistant AI: control room

Runs locally (no paid cloud) · never needed to run the exam

 Explains decisions	Plain-language reasons officials can approve
 Repair estimate	How long until the centre recovers
 Seat planner	Moves candidates to free seats (optimiser)
 Risk ranking	Risky centres before exam day; repeat problems
 Drafts	Messages and reports in Hindi and English

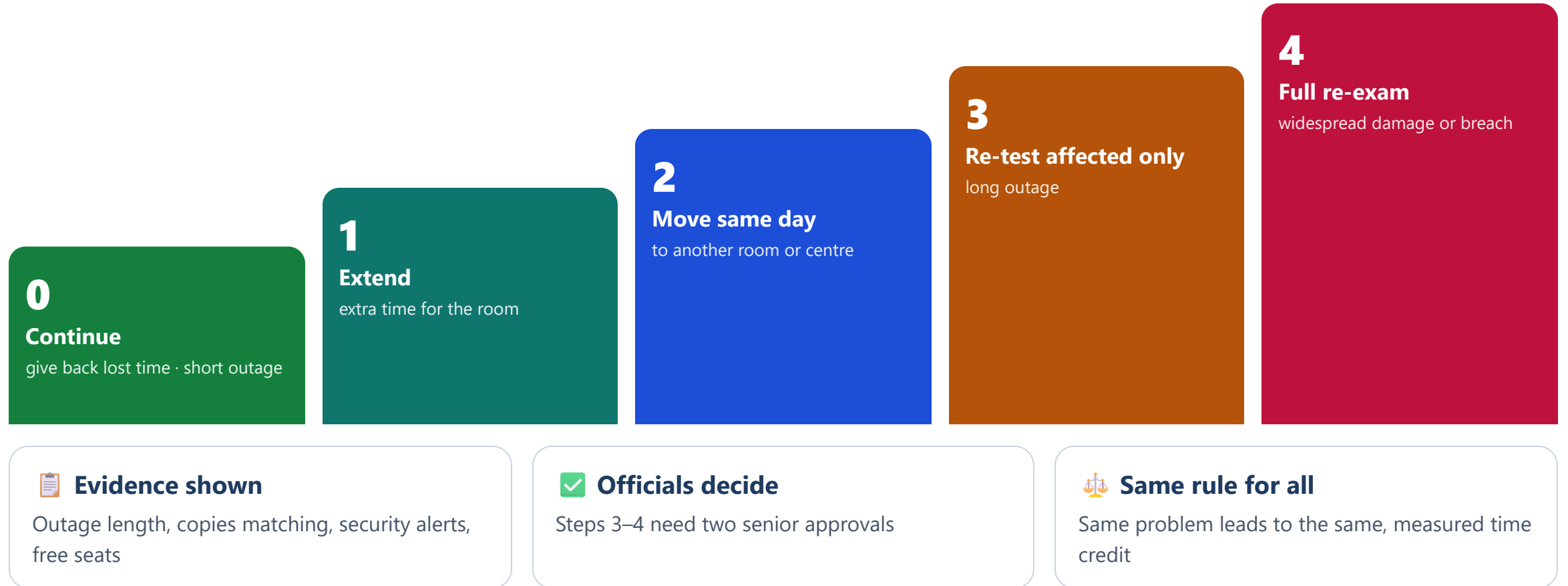
 Every AI output shows its reasons · a rule-based backup for every model · officials approve important decisions · no personal data goes to AI

Everyone sees their part. Nobody can secretly change it.



~80 attack methods studied across 9 attacker types, from candidates to insiders, each with a prevention, detection and response.

The 5-step decision guide: no needless re-exams



Ordinary hardware. Huge safety margins.

99.6%

of a 100 Mbps centre network left free at peak

0.1%

of a 10 Mbps internet line used to sync

≈ ₹5

hardware cost per candidate per exam

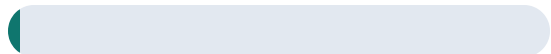
1:10,000

chance both centre servers fail together



Centre network at peak (300 candidates)

ASTRA traffic



0.4%

Free



99.6%

$300 \times 300 \text{ saves} \times 150 \text{ bytes} = \mathbf{13.5 \text{ MB per exam}}$ · state scale (5 lakh candidates) $\approx \mathbf{13,900 \text{ records/s}}$, $\sim 7\times$ under typical database limits



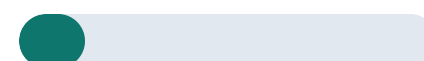
One avoided cancellation pays for 6 centres

Cancelled shift ($300 \times ₹3,100$)



₹9.3 L

ASTRA hardware per centre



₹1.46 L

2 servers + 2 UPS + switch + 4G router · reused for 100 exam days · **₹0 software licences**

Built entirely on **free, open-source technology**

Frontend

 React

 TypeScript

 Vite

 Tailwind CSS

 Offline web app
(PWA)

 WebRTC (CCTV view)

 Recharts

Backend

 Python 3.12

 FastAPI

 WebSockets + REST

 Ed25519 · AES-256 ·
TLS

 Two-step login
(TOTP)

 Encrypted links (TLS)

ML / AI

 LightGBM

 scikit-learn

 OpenCV (vision)

 Laya (fast AI)

 Ollama (local LLM)

 OR-Tools

 pandas

 NumPy

Infra

 SQLite

 PostgreSQL

 Debian kiosk image

 VirtualBox

 FFmpeg

 MediaMTX (CCTV)

 GitHub Pages

What changes for candidates and exam authorities

Today

- Power cut or server crash → shift cancelled
- Hours of silence and rumours
- Lost time guessed, re-test weeks later
- Evidence collected afterwards

With ASTRA

- Pause, then continue: **no answers or minutes lost**
- Official update in **under 30 seconds**
- **Only affected** candidates re-tested
- Tamper-proof record from the first second

Measure	Today	ASTRA target
Answers lost after a failure	Not tracked	0
Time until candidates are told	Up to hours	< 30 s
Time to continue after a PC crash	Often never (re-test)	< 90 s
Same centre failing again	Not noticed	Flagged before the next exam

Digital India

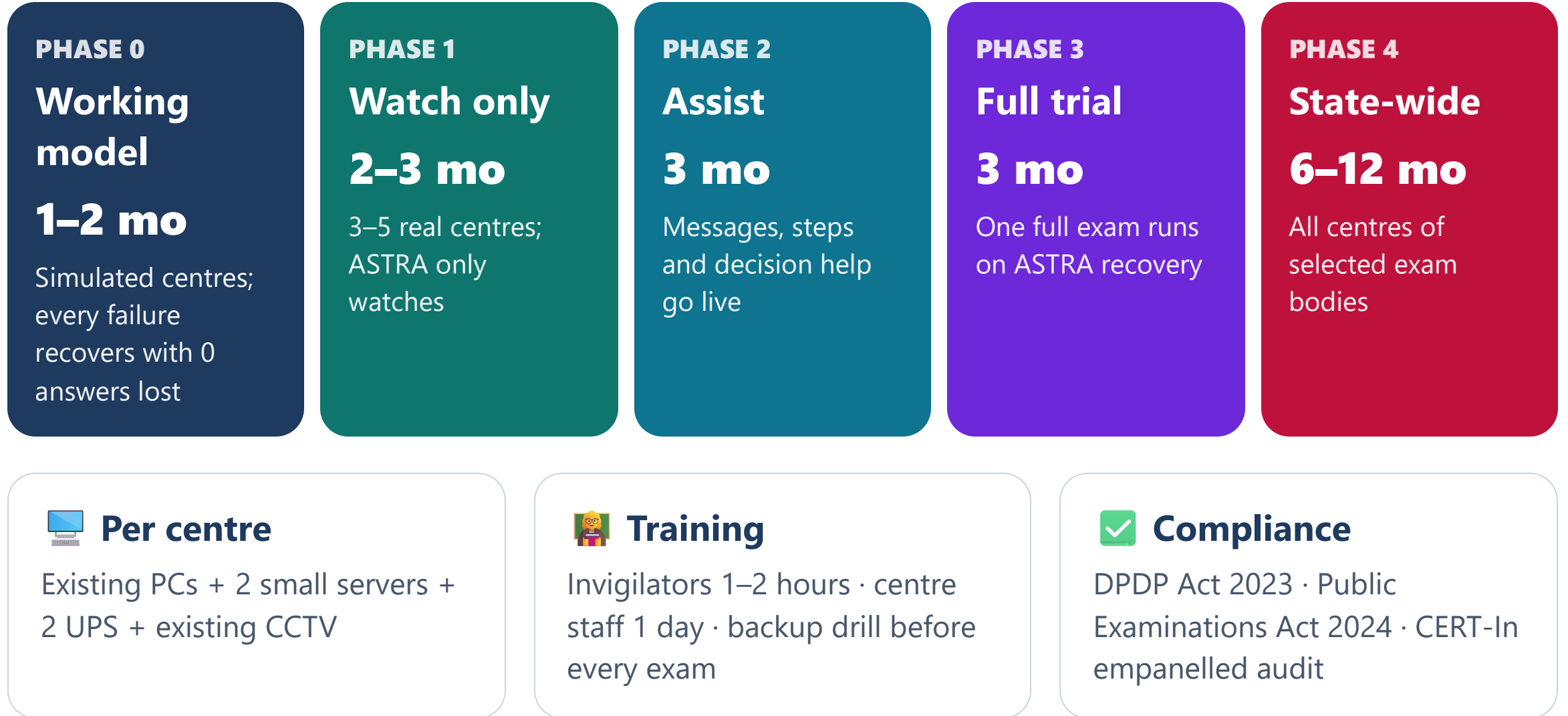
NEP 2020

Viksit Bharat 2047

Good Governance

Trust in digital public services

No sudden switchover: trust earned step by step



Failures pause. **Exams don't.**

★ New core idea

Every answer in 3 places; continue on any PC with exact time

★ Safe, fast AI

Predicts before failure; never blocks the exam; humans decide

★ Proven affordable

₹0 licences · $\approx$ ₹5 per candidate · 99.6% network headroom



Live prototype



GitHub repository

yash-kasera.github.io/ASTRA/prototype

github.com/yash-kasera/ASTRA

Team Marauders